

Akash Vani

Max Planck Institute for Astrophysics, Garching, Germany

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Galaxy Evolution & Quenching | The Early Universe | Numerical Simulations

Education

Present | **Ph.D. in Physics** | Max Planck Institute for Astrophysics (IMPRS-A) & Ludwig Maximilian University of Munich, Germany

Advisors: Prof. Dr. Volker Springel, Prof. Dr. Guinevere Kauffmann, MPA, Garching

2022 | **M.Sc. in Physics** | Heidelberg University, Germany

Thesis: *Properties of stellar populations and white dwarfs in a semi-analytic Milky Way disk model based on Gaia DR3*

Advisor: Prof. Dr. Andreas Just, Astronomisches Rechen-Institut (ARI), ZAH, Heidelberg

2019 | **B.Sc. in Physics** | Savitribai Phule Pune University, India

Thesis: *Photometric calibration of UVIT* (UV telescope on AstroSat)

Advisor: Prof. Dr. Shyam Tandon, Inter-University Centre for Astronomy and Astrophysics (IUCAA), Pune

Appointments

- Sep 2023 – Present | Doctoral Researcher, **Max Planck Institute for Astrophysics**, Garching
- Jun 2026 – Sep 2026 | JSPS Fellow & Visiting Doctoral Researcher, **University of Tokyo**, Tokyo
- Nov 2022 – Aug 2023 | Research Intern, **Max Planck Institute for Astrophysics**, Garching
- Oct 2022 – Sep 2023 | Predoctoral Student, **University of Heidelberg**, Heidelberg
- Oct 2021 – Oct 2022 | Student Assistant, **German Cancer Research Center (DKFZ)**, Heidelberg
- Sep 2021 – Sep 2022 | Master's Thesis Researcher, **Astronomisches Rechen-Institut, ZAH**, Heidelberg
- Aug 2019 – Oct 2020 | Visiting Student Researcher, **Tata Institute of Fundamental Research**, Mumbai
- Jun 2019 – Jul 2019 | Quantum Information and Technology Summer Fellow, **IISER Kolkata**, Kolkata
- Jun 2018 – Apr 2019 | Bachelor's Thesis Student, **Inter-University Centre for Astronomy and Astrophysics (IUCAA)**, Pune

Fellowships & Awards

- 2026 | JSPS Fellowship hosted at the University of Tokyo
- 2023 | Max Planck Society Doctoral Fellowship
- 2022 | MPA Research internship

Publications

Total: 10 | Refereed: 4 | Submitted: 3 | In preparation: 3 | Citations: 86

1. **Vani, A.**, Ayromlou, R., Kauffmann, G., & Springel, V., 'Galaxies in the first two billion years: Earlier formation and quenching with surface-density modulated star formation and feedback', submitted (2026), [arXiv:2609.12060](https://arxiv.org/abs/2609.12060).
2. Sysoliatina, K*. & **Vani, A.***, 'The full Gaia colour-magnitude diagram with white dwarfs reveals a declining Milky Way star formation history', submitted (2026). *Joint lead authors
3. Golovin, A., Reffert, S., **Vani, A.**, 'The Sixth Catalogue of Nearby Stars (CNS6)', submitted (2026).
4. Pascalau, R., et al. (incl. **Vani, A.**), 'When relics were made: vigorous stellar rotation and low dark-matter content in the massive ultra-compact galaxy GS-9209 at $z = 4.66$ ', MNRAS (2026) [2026MNRAS.547ag210P](https://arxiv.org/abs/2026MNRAS.547ag210P).
5. **Vani, A.**, Ayromlou, R., Kauffmann, G., & Springel, V., 'Probing galaxy evolution from $z = 0$ to $z \simeq 10$ through galaxy scaling relations in three L-Galaxies flavours', MNRAS (2025). [2025MNRAS.536..777V](https://arxiv.org/abs/2025MNRAS.536..777V)
6. Golovin, A., Reffert, S., **Vani, A.**, Bastian, U., Jordan, S., & Just, A., 'Identification of 29 new nearby white dwarfs using Gaia DR3', A&A (2024). [2024A&A...683A..33G](https://arxiv.org/abs/2024A&A...683A..33G)
7. Golovin, A., Reffert, S., Just, A., Jordan, S., **Vani, A.**, & Jahreiß, H., 'The Fifth Catalogue of Nearby Stars (CNS5)', A&A (2023). [2023A&A...670A..19G](https://arxiv.org/abs/2023A&A...670A..19G)

Research Experience

Galaxy Formation & Evolution

- Optimisation, statistical inference, and deep-learning applications for galaxy-evolution models.
Collaboration: V. Springel (MPA, Garching); R. Ayromlou (AlfA, Bonn); N. Yoshida (UTokyo & IPMU, Tokyo)
- JWST NIRCам/NIRSpec study of galaxy interactions and evolution in a protocluster environment.
Collaboration: R. Maiolino (UCambridge & KICC)
- Development of a new and efficient galaxy-evolution framework with advanced physical prescriptions for early-Universe star formation, feedback, and model calibration.
Collaboration: V. Springel, G. Kauffmann (MPA, Garching); R. Ayromlou (AlfA, Bonn)
- Diagnostic study and stress testing of galaxy-evolution models across cosmic time.
Collaboration: G. Kauffmann, V. Springel (MPA, Garching); R. Ayromlou (ITA, Heidelberg)

Milky Way Stellar Populations & White Dwarfs

- Milky Way modelling with main-sequence, white-dwarf, and substellar population synthesis.
Collaboration: K. Sysoliatina, A. Just, S. Jordan (ARI, Heidelberg)
- Census and characterisation of stars within 25 pc using *Gaia* and complementary space- and ground-based surveys.
Collaboration: A. Golovin, S. Reffert, A. Just, S. Jordan (LSW & ARI, Heidelberg)

Competencies

Programming	Python (AstroPy, NumPy, SciPy, Matplotlib, scikit-learn, Optuna, Pytorch), C/C++, Bash
Statistics	MCMC, Bayesian analysis, machine & deep learning
HPC	SLURM, SGE, large-scale parallel workflows
Tools	Git/GitHub, L ^A T _E X, HTML, DS9, TOPCAT, Linux/Unix, MS Office, Adobe Photoshop & Illustrator
Languages	English, German, Marathi, Konkani, Hindi
Software	One of the main developers of L-GALAXIES (cosmological galaxy evolution) and CALOMERA (chemo-dynamical Milky Way disc)

Talks & Posters

Selected from 43 talks (12 invited), 2020–2026.

- ‘*Grow fast, die young: the early universe’s galaxy formation recipe*’ — AG Meeting, U. Osaka, U. Tokyo, Kavli IPMU, CEA Saclay (2026)
- ‘*Quenching that shapes the first billion years*’ — U. Bonn, ICRR Tokyo (2026); TIFR Mumbai (2025)
- ‘*Probing galaxy number density, sizes, and surface densities in L-Galaxies*’ — L-Galaxies Seminar Series (2025)
- ‘*Properties of quenched galaxies in the L-Galaxies framework across cosmic time*’ — Multiscale Environment of AGN (2025)
- ‘*Debugging galaxy evolution across cosmic epochs using L-Galaxies*’ — U. Zurich (2025), STScI Spring Symposium, Baltimore (2024), IAU General Assembly, Cape Town (2024)

Professional & Outreach Activities

Leadership & Service

- Feb 2025 – Present | Mentor, MPA mentoring programme for underrepresented groups
- May 2024 – Present | PhD Representative, MPA, Germany
- Nov 2023 – Present | Social-media Working Group, PhDnet@MPG, Germany
- Oct 2025 – Nov 2025 | Local Organising Committee, AGN conference, Ringberg, MPA
- Sep 2024 – Nov 2024 | Chair of SOC, 18th IMPRS Symposium, Garching, Germany
- Sep 2021 – Oct 2023 | International Student Buddy, Heidelberg University, Germany

Outreach

- Research highlight of the month (Feb 2025), featured on the [MPA highlight webpage](#)
- Public-outreach talk on recent developments in astronomy for school (Aug 2023, Goa) and undergraduate students (Jan 2026, Goa).
- Science demonstrations at open days – high school (Goa, 2014–16), university (Pune, 2016–19) – and a national competition (Mumbai, 2016)